

# Benchmarking of NQCH's quantum computer

May 21, 2026

## 1. Report of Changes

**Platform:** sinq20

**Calibration-id:** d7f8a21692dc9acaa78c0ecabd13a51101707b4c

**Calibration date:** 2026-05-13 06:56:33

**Calibration note:** Merge pull request #17 from qiboteam/sqcal-280426

Sqcal 280426

**Experiment-id:** 20260520212203

**Experiment date:** 2026-05-20 21:22:04

**Experiment note:** temporary note!!!

**Platform:** sinq20

**Calibration-id:** 4dc4082f38a53222b3956c22202d32a520d4bc78

**Calibration date:** 2026-01-15 02:03:03

**Calibration note:** chore(sinq20): 2q gates cal 0-1, 0-3, 2-3, 3-4, 1-4, 4-5, 4-9, 3-8, 8-9

**Experiment-id:** 20260118161538

**Experiment date:** 2026-01-18 16:15:38

**Experiment note:** temporary note!!!

## 2. Version Comparison

| Library      | Version | Library | Version |
|--------------|---------|---------|---------|
| qibo         | 0.2.23  | numpy   | 2.4.4   |
| qibolab      | 0.2.9   | qibocal | 0.2.5   |
| matplotlib   | 3.10.8  | scipy   | 1.17.1  |
| scikit-learn | 1.8.0   | pandas  | 2.3.3   |
| networkx     | 3.6.1   | sympy   | 1.14.0  |
| torch        | 2.11.0  |         |         |

| Library      | Version | Library | Version |
|--------------|---------|---------|---------|
| qibo         | 0.2.22  | numpy   | 2.2.6   |
| qibolab      | 0.2.9   | qibocal | 0.2.4   |
| matplotlib   | 3.10.3  | scipy   | 1.15.3  |
| scikit-learn | 1.6.1   | pandas  | 2.2.3   |
| networkx     | 3.4.2   | sympy   | 1.14.0  |
| torch        | 2.7.0   |         |         |

## 3. One and two qubit fidelities

The single qubit fidelity is obtained via Randomized-Benchmarking. The two-qubit fidelity is the "Bell-state fidelity".





## 8. Mermin

Mermin's algorithm for 3 qubits.

| Runtime    | Qubits  | Max   |
|------------|---------|-------|
| 239.81 sec | 0, 8, 3 | 3.338 |



| Runtime  | Qubits  | Max    |
|----------|---------|--------|
| 1.02 sec | 0, 2, 3 | -3.316 |



## 9. Grover - 2 qubits

Grover's algorithm for 2 qubits executed on sinq20 backend with 1000 shots per circuit. We measure the success rate of finding the target state '11' for each pair of qubits in  $[[0, 3]]$ .

| Runtime  | Qubits | Max prob |
|----------|--------|----------|
| 3.97 sec | 0, 3   | 0.882    |



| Runtime  | Qubits | Max prob |
|----------|--------|----------|
| 2.36 sec | 2, 3   | 0.89     |



## 10. Grover - 3 qubits

Grover's algorithm for 3 qubits executed on `siq20` backend with 1000 shots per circuit. We measure the success rate of finding the target state '111' for each pair of qubits in `[[0, 3], [8, 3]]`.

| Runtime  | Qubits  | Max prob |
|----------|---------|----------|
| 3.23 sec | 8, 0, 3 | 0.292    |



| Runtime  | Qubits  | Max prob |
|----------|---------|----------|
| 2.43 sec | 0, 2, 3 | 0.385    |



## 11. GHZ state preparation

GHZ circuit with 3 qubits executed on `siq20` backend with 1000 shots.

| Runtime  | Qubits  | Success |
|----------|---------|---------|
| 3.06 sec | 0, 3, 8 | 0.897   |



| Runtime  | Qubits  | Success |
|----------|---------|---------|
| 3.95 sec | 0, 3, 2 | 0.910   |

